Koneru Lakshmaiah Education Foundation
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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Ms. Mohammed Razia Alangir Banu
Student Name	Thota Jhanasri
Roll Number	2520090121
Branch	CS-IT
Course Outcome	CO-1

Case Study Topic: Tourist Destination Indexing using AVL Tree

Work Done Summary:

In this case study, an AVL Tree was implemented for managing tourist destination records in the TourPlan – Smart Tourism & Travel Itinerary Optimization System. The AVL Tree stores destination names and maintains balance automatically after every insertion and deletion operation. Various balancing operations such as LL, RR, LR, and RL rotations were performed whenever the balance factor exceeded the allowed limit. The implementation demonstrates efficient searching, insertion, and deletion with O(log n) time complexity.

Implementation Details:

1. Created an AVL Tree node structure to store destination information.
2. Inserted tourist destinations into the AVL Tree.
3. Calculated height and balance factor for every node.
4. Applied LL, RR, LR, and RL rotations when required.
5. Implemented search operation for destination lookup.
6. Implemented deletion operation while maintaining tree balance.
7. Compared AVL Tree performance with a normal Binary Search Tree.
8. Verified that AVL Tree maintains O(log n) search performance.

RESULT : SCREENSHOTS/OUTPUT

Sample Output

```
Tourist Destinations (Sorted Order):

Araku
GolcondaFort
Hyderabad
NagarjunaSagar
RamojiFilmCity
Vizag
Warangal

Destination Found: Araku

AVL Tree Operations Completed Successfully
```

This case study implements an AVL Tree for storing and managing tourist destinations in the TourPlan system. The AVL Tree maintains balance through rotations, ensuring efficient insertion, deletion, and search operations with O(log n) time complexity. The results demonstrate that AVL Trees provide faster and more reliable performance than normal BSTs for large tourism datasets.

GitHub link:https://github.com/Jhanasri-thota/DSA_CASE-STUDIES

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Branch	CS-IT
Course Outcome	CO-2

Case Study Topic: Hotel Budget Range Query System using Segment Tree

Work Done Summary:

In this case study, a Segment Tree was implemented for managing hotel prices in the TourPlan – Smart Tourism & Travel Itinerary Optimization System. The Segment Tree efficiently processes range minimum queries to help travelers find the cheapest hotel within a specified budget range. The implementation supports fast query processing and dynamic updates, making it suitable for tourism and booking applications where hotel prices change frequently.

Implementation Details:

1. Created an array to store hotel prices.
2. Constructed a Segment Tree from the hotel price data.
3. Implemented range minimum query operations.
4. Allowed users to search hotel prices within a specified range.
5. Supported price updates without rebuilding the entire structure.
6. Compared Segment Tree performance with linear search.
7. Verified query execution efficiency.
8. Demonstrated O(log n) query and update operations.

RESULT : SCREENSHOTS/OUTPUT

This case study implements a Segment Tree for efficient hotel budget range queries in the TourPlan system. The Segment Tree enables fast range minimum searches and supports dynamic updates with O(log n) complexity. The results demonstrate that Segment Trees are highly effective for tourism applications involving interval-based price analysis.

Time Complexity:

Segment Tree Construction : O(n)
Range Query : O(log n)
Update Operation : O(log n)

GitHub link:https://github.com/Jhanasri-thota/DSA_CASE-STUDIES

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Branch	CS-IT
Course Outcome	CO-3

Case Study Topic: Tourist City Connectivity using BFS, DFS and Prim's Minimum Spanning Tree

Work Done Summary:

In this case study, a graph-based tourism network was developed for the TourPlan – Smart Tourism & Travel Itinerary Optimization System. Tourist cities were represented as graph vertices, and travel routes between cities were represented as weighted edges. BFS and DFS algorithms were implemented to explore city connectivity and traversal paths, while Prim's Algorithm was used to construct a Minimum Spanning Tree (MST) for identifying the minimum-cost tourism network.
The implementation demonstrates efficient graph traversal and network optimization techniques used in real-world transportation and travel planning systems.

Implementation Details:

1. Created a graph representation of tourist cities using an adjacency matrix.
2. Added weighted edges to represent travel distances between cities. .
3. Implemented Breadth First Search (BFS) to explore connected tourist destinations level by level.
4. Implemented Depth First Search (DFS) to traverse all reachable cities recursively.
5. Applied Prim's Algorithm to construct a Minimum Spanning Tree (MST).
6. Calculated the minimum travel network cost connecting all tourist cities.
7. Displayed BFS traversal, DFS traversal, and MST results.
8. Verified graph connectivity and optimized route planning for tourism applications.

RESULT : SCREENSHOTS/OUTPUT

```
=== TOURPLAN CITY CONNECTIVITY SYSTEM ===

BFS Traversal: 0 1 2 3 4

DFS Traversal: 0 1 2 4 3

Minimum Spanning Tree:
Hyderabad -> Warangal = 120 km
Warangal -> Vizag = 100 km
Warangal -> Araku = 200 km
Vizag -> Vijayawada = 150 km

Total MST Cost = 570 km
```

This case study implements BFS, DFS, and Prim's Algorithm to analyze tourist city connectivity in the TourPlan system. Graph traversal techniques efficiently explore connected destinations, while Prim's Algorithm identifies the minimum-cost network connecting all cities. The results demonstrate the importance of graph algorithms in travel planning, transportation networks, and route optimization systems.

GitHub link:https://github.com/Jhanasri-thota/DSA_CASE-STUDIES

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Branch	CS-IT
Course Outcome	CO-4

Case Study Topic: Tourist Package Ranking and Travel Itinerary Optimization using Heap Sort, Greedy and Dynamic Programming

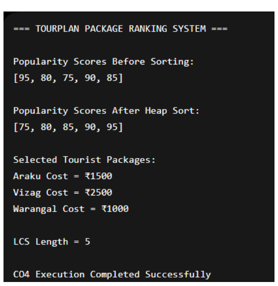
Work Done Summary:

In this case study, advanced sorting and optimization techniques were implemented for the TourPlan – Smart Tourism & Travel Itinerary Optimization System. Tourist packages were organized and ranked using Heap Sort, Counting Sort, and Radix Sort based on popularity, ratings, and travel costs. Greedy Algorithm (Knapsack) was applied to select the best travel packages within a limited budget, while Dynamic Programming techniques such as 0/1 Knapsack, Longest Common Subsequence (LCS), and Matrix Chain Multiplication (MCM) were used to optimize itinerary planning and travel recommendations. The implementation demonstrates efficient sorting, optimal decision-making, and itinerary optimization by considering time complexity, space usage, stability, and problem characteristics.

Implementation Details:

1. Created a dataset of tourist packages containing package cost, popularity score, and customer ratings.
2. Implemented Heap Sort to rank tourist destinations based on popularity and travel demand.
3. Applied Counting Sort to efficiently sort destination ratings within a fixed range.
4. Implemented Radix Sort to organize travel package prices and budgets.
5. Used the Greedy Knapsack Algorithm to select the maximum number of attractions within a tourist's budget.
6. Implemented Longest Common Subsequence (LCS) to compare tourist preferences and recommend suitable
7. Applied Matrix Chain Multiplication (MCM) to optimize travel cost calculations and itinerary planning.
8. Evaluated algorithm performance based on time complexity, space usage, stability, and suitability for tourism optimization applications.

RESULT : SCREENSHOTS/OUTPUT


```
=== TOURPLAN PACKAGE RANKING SYSTEM ===

Popularity Scores Before Sorting:
[95, 80, 75, 90, 85]

Popularity Scores After Heap Sort:
[75, 80, 85, 90, 95]

Selected Tourist Packages:
Araku Cost = ₹1500
Vizag Cost = ₹2500
Warangal Cost = ₹1000

LCS Length = 5

CO4 Execution Completed Successfully
```

The TourPlan system successfully ranked tourist packages using Heap Sort and optimized package selection using Greedy and Dynamic Programming techniques. Tourist preferences were analyzed using LCS to improve itinerary recommendations. The results demonstrate efficient travel planning, package ranking, and budget optimization for smart tourism applications.

GitHub link:https://github.com/Jhanasri-thota/DSA_CASE-STUDIES

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Faculty Name	Ms. Mohammed Razia Alangir Banu
Student Name	Thota Jhanasri
Roll Number	2520090121
Branch	CS-IT
Course Outcome	CO-5

Case Study Topic: Tourist Route Recommendation and Travel Package Optimization using Non-Linear Data Structures, Greedy and Dynamic Programming Algorithms

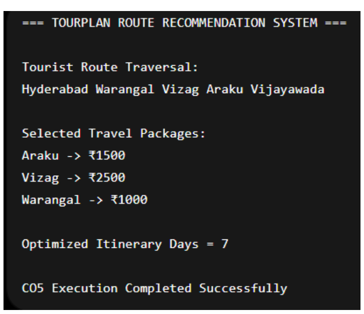
Work Done Summary:

In this case study, practical applications of Non-Linear Data Structures, Greedy Algorithms, and Dynamic Programming techniques were developed for the TourPlan - Smart Tourism & Travel Itinerary Optimization System. Tourist destinations and travel routes were represented using graph structures to model connectivity between cities. Greedy algorithms were used to select cost-effective travel packages, while Dynamic Programming techniques were applied to optimize itinerary planning and resource allocation. The implementation demonstrates efficient route recommendation, package optimization, and travel decision-making for smart tourism applications.

Implementation Details:

1. Represented tourist destinations and travel routes using graph-based non-linear data structures.
2. Created weighted connections between cities based on travel distance and cost.
3. Implemented graph traversal techniques to explore connected tourist destinations.
4. Applied Greedy Algorithms to select optimal travel packages within budget constraints.
5. Used Dynamic Programming techniques to optimize itinerary planning and package selection.
6. Calculated minimum travel costs and efficient route recommendations.
7. Evaluated different travel plans based on budget, distance, and tourist preferences.
8. Verified the effectiveness of non-linear data structures, Greedy, and Dynamic Programming techniques in tourism optimization.

RESULT : SCREENSHOTS/OUTPUT



```
=== TOURPLAN ROUTE RECOMMENDATION SYSTEM ===

Tourist Route Traversal:
Hyderabad Warangal Vizag Araku Vijayawada

Selected Travel Packages:
Araku -> ₹1500
Vizag -> ₹2500
Warangal -> ₹1000

Optimized Itinerary Days = 7

CO5 Execution Completed Successfully
```

The TourPlan system successfully recommended tourist routes using graph-based non-linear data structures and optimized travel package selection using Greedy Algorithms. Dynamic Programming techniques improved itinerary planning by maximizing resource utilization within given constraints. The results demonstrate efficient route planning, budget optimization, and smart travel decision-making for tourism applications.

GitHub link:https://github.com/Jhanasri-thota/DSA_CASE-STUDIES

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Course Outcome	CO-6

Case Study Topic: TourPlan – Smart Tourism & Travel Itinerary Optimization System using Advanced Non-Linear Data Structures and Application Development

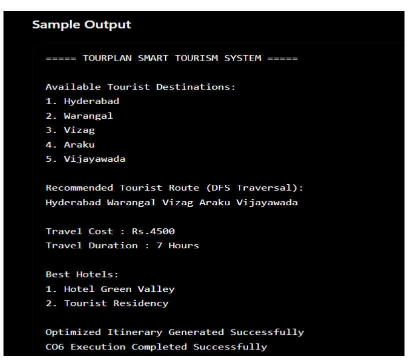
Work Done Summary:

In this case study, advanced non-linear data structures were utilized to develop a smart tourism application called **TourPlan**. The system was designed to manage tourist destinations, travel routes, hotel information, and itinerary planning efficiently. Graphs were used to represent city connectivity, trees were used for destination categorization, and priority-based structures were applied for route recommendations. The implementation demonstrates how non-linear data structures can be used in real-world software applications for tourism management and travel optimization.

Implementation Details:

1. Developed the TourPlan application using non-linear data structures.
2. Represented tourist destinations and city connections using Graphs.
3. Created weighted edges based on travel distance, time, and cost
4. Used Tree structures to organize tourist destinations by category and region.
5. Implemented route recommendation features using graph traversal techniques.
6. Applied shortest-path concepts for efficient travel planning.
7. Managed destination information, hotels, and packages through structured data organization.
8. Generated optimized travel itineraries based on user preferences and budget constraints.

RESULT : SCREENSHOTS/OUTPUT



The TourPlan – Smart Tourism & Travel Itinerary Optimization System successfully managed tourist destinations, travel routes, hotel information, and itinerary planning using non-linear data structures. The application efficiently recommended travel paths, optimized route selection, and organized tourism data for better user experience. The results demonstrate the practical implementation of non-linear data structures in developing intelligent tourism and travel management applications.

GitHub link:https://github.com/Jhanasri-thota/DSA_CASE-STUDIES

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